Auteur: Odia Kibor
Studierichting: Informatica
Oefeningenreeks 2E nr. 4.7

$$\lim_{x\to\infty}\left(\frac{ax+b}{ax+c}\right)^x$$
$$=\lim_{x\to\infty}\left(\frac{(ax+c)+(b-c)}{ax+c}\right)^x$$
$$=\lim_{x\to\infty}\left(1+\frac{b-c}{ax+c}\right)^x$$
Stel $\frac{1}{u}=\frac{b-c}{ax+c}\Rightarrow u=\frac{ax+c}{b-c}\Rightarrow x=\frac{u(b-c)-c}{a}$
$$=\lim_{u\to\infty}\left(1+\frac{1}{u}\right)^{\frac{u(b-c)-c}{a}}$$
$$=\lim_{u\to\infty}\left(1+\frac{1}{u}\right)^{\frac{u(b-c)}{a}-\frac{c}{a}}$$
$$=\lim_{u\to\infty}\left(1+\frac{1}{u}\right)^{\frac{u(b-c)}{a}}\cdot\left(1+\frac{1}{u}\right)^{-\frac{c}{a}}$$
$$=\lim_{u\to\infty}\left(\left(1+\frac{1}{u}\right)^u\right)^{\frac{b-c}{a}}\cdot\lim_{u\to\infty}\left(1+\frac{1}{u}\right)^{-\frac{c}{a}}$$
$$=e^{\frac{b-c}{a}}\cdot 1$$
$$=e^{\frac{b-c}{a}}$$

References